Comparing Fine-Tuning Strategies for Bribri Text-to-Speech in an Extreme Low-Resource Setting

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Abstract

Modern text-to-speech (TTS) has advanced quickly for high-resource languages, but it remains out of reach for most Indigenous languages, which simply do not have the large aligned corpora these systems assume. We study Bribri, a tonal Chibchan language of Costa Rica with roughly 7,000 speakers, as a case of extreme low-resource TTS. Using about 25 minutes of speech (449/56/56 clips) pooled from four native speakers, we compare three transfer-learning strategies on identical test sentences: fine-tuning MMS-TTS (VITS) from Cabécar, a close sister language; from Spanish, a distant high-resource one; and LoRA fine-tuning of Spark-TTS, an LLM-based model. By Mel Cepstral Distortion (MCD) the VITS systems scored best (Cabécar 3.06, Spanish 3.45 dB) against a 7.18 dB baseline, while Spark-TTS scored 6.44 dB. But a native speaker's Mean Opinion Score (MOS) reversed the ranking: Spark-TTS rated highest (2.75), tying Cabécar VITS (2.55) and beating both Spanish systems (2.10). Our central finding is that fine-tuning mattered far more than sister-language proximity. The metric disagreement supports our argument that the MCD gap is an artifact of Spark-TTS's lack of speaker conditioning, and that spectral distance alone cannot rank these architectures for tonal low-resource TTS.

1 Introduction

Neural text-to-speech (TTS) now produces near-natural speech in high-resource languages, yet the benefits of this are spread very unevenly. These systems depend on large aligned speech corpora, stable orthographies, and enough speaker data to learn consistent pronunciations, and those assumptions routinely fail for Indigenous and low-resource languages. So the same technologies that make English, Spanish, or Mandarin more accessible through voice interfaces end up leaving many smaller languages almost completely absent from digital spaces.

We study Bribri, an Indigenous Chibchan language spoken in Costa Rica by roughly 7,000 people, as a case of this problem. Bribri is also tonal, which only adds to the difficulty, because a synthesizer has to learn not just how the segments are pronounced but also the pitch distinctions that the tone diacritics carry. The practical motivation here is language revitalization: a working Bribri voice could support reading apps, pronunciation tools, classroom materials, and other accessibility resources. The technical conditions, however, are severe, since the entire accessible corpus is only about 25 minutes of audio, far below what neural TTS would normally need.

We compare three transfer-learning strategies. The first fine-tunes *facebook/mms-tts-cjp*, a VITS-based MMS-TTS model pretrained on Cabécar, the closest sister language to Bribri. The second fine-tunes *facebook/mms-tts-spa*, pretrained on Spanish—phonetically more distant but backed by far more pretraining audio. The third LoRA-fine-tunes *unsloth/Spark-TTS-0.5B*, a recent LLM-based system that predicts discrete speech tokens and decodes them to waveform.

Our main research question is how these approaches behave for Bribri under such an extreme low-resource constraint, and in particular whether how closely the base language is related to Bribri is really the deciding factor. Our headline result is that it is not: fine-tuning improves both base models enormously, but the choice between a close sister language and a distant high-resource one barely moves the objective metric at all. A second contribution is methodological. We show that the apparent gap between the VITS and Spark-TTS architectures on Mel Cepstral Distortion is confounded by the absence of speaker conditioning in our Spark-TTS setup, and we argue that objective spectral metrics cannot, on their own, adjudicate these architectures for tonal low-resource speech.

Prior work frames these comparisons. VITS introduced an end-to-end TTS model combining a conditional variational autoencoder, normalizing flows, adversarial training, and a stochastic duration predictor (Kim et al., 2021). Meta's Massively Multilingual Speech (MMS) project scaled speech technology past 1,000 languages and released per-language VITS synthesis checkpoints, including the Cabécar and Spanish models we adapt (Pratap et al., 2024). Spark-TTS represents a newer, LLM-based line of speech synthesis using single-stream decoupled speech tokens built on

Qwen2.5 with a BiCodec decoder (Wang et al., 2025); LoRA enables parameter-efficient adaptation of such models (Hu et al., 2022). To our knowledge, this is among the first applications of Spark-TTS to an Indigenous language of the Americas.

2 Methodology

2.1 Data

The Bribri speech data was provided by Prof. Rolando Coto-Solano and collected through prior work with Bribri speakers; in his words it is "about all there really is." The accessible corpus totals roughly 25 minutes of 16 kHz mono WAV audio with CSV transcripts, split into 449 training, 56 validation, and 56 test clips. Three constraints define the setting. First, 25 minutes is well below the practical minimum for training or even fine-tuning neural TTS. Second, the recordings come from four native speakers but were pooled into a single-speaker training target, which uses all available data at the cost of a stable speaker identity. Third, Bribri tone and orthography require care: we apply NFD Unicode normalization so that tone diacritics become separate combining-mark codepoints, and (for the VITS models) extend the tokenizer with five such marks—U+0300, U+0301, U+0302, U+0303, and U+0308—resizing the text-encoder embeddings while preserving pretrained rows.

2.2 MMS-TTS (VITS)

Our first two systems fine-tune MMS-TTS checkpoints. VITS maps text to waveform end-to-end through a latent acoustic representation, monotonic alignment search for duration, and adversarially trained generation, which makes it powerful but unstable when data is scarce (Kim et al., 2021). We chose Cabécar (*cjp*) because it is closely related to Bribri, and Spanish (*spa*) as a contrasting, more distant but higher-resource base, isolating the effect of base-language proximity. Both base checkpoints derive from a New-Testament pretraining corpus.

Fine-tuning was not stable out of the box. Our initial run (100 epochs, learning rate 2e-5, discriminator weight 3) suffered generator collapse, producing naturalistic but phonetically wrong speech. We recovered by shortening training and shifting the loss heavily toward spectral reconstruction (Table 1). The final configuration trained for 30 epochs at learning rate 1e-5 with a mel-loss weight of 65 against a discriminator weight of 0.25. This trade-off recurs throughout our results: more mel weight sharpens phonemes but adds artifacts, while more discriminator weight sounds more human but degrades intelligibility.

Parameter	Failed	Final
Epochs	100	30
Learning rate	2e-5	1e-5
weight_mel	35	65
weight_disc	3	0.25

Table 1: VITS fine-tuning hyperparameters before and after recovering from generator collapse. The final recipe trades adversarial pressure for spectral-reconstruction pressure.

2.3 Spark-TTS

Our third system fine-tunes *unsloth/Spark-TTS-0.5B*, released in March 2025. Spark-TTS decouples linguistic and acoustic modeling: a Qwen2.5-0.5B language model predicts discrete speech tokens, and a separately trained BiCodec decoder converts them to waveform (Wang et al., 2025). This is attractive for low-resource work because the decoder already contains general acoustic knowledge, so adaptation need not relearn waveform generation from 25 minutes of Bribri; because training is next-token cross-entropy rather than adversarial, it avoids GAN instability; and because the underlying LLM was exposed to Mandarin, a tonal language, it may—speculatively—carry a useful tonal prior. We fine-tuned with LoRA (rank 128, α=128) on all attention and MLP projection matrices (Hu et al., 2022), training for 60 steps at learning rate 2e-4 with an effective batch size of 8 (per-device batch 2, gradient accumulation 4) in bf16. Training took roughly 70 seconds on a Colab L4 GPU.

2.4 Evaluation

All three systems synthesized the same 20 Bribri test sentences, selected deterministically by NFD length and stride sampling so that the same items appear across conditions for per-sentence comparison; the same sentences were also produced by a human speaker as ground truth. Our objective metric is Mel Cepstral Distortion (MCD), the spectral distance between synthesized and reference audio, where lower is better. MCD gives a consistent number across systems but measures spectral similarity only—not intelligibility, tone correctness, or phonetic faithfulness—so we treat it as diagnostic rather than decisive. To capture perceived quality, we complement MCD with Mean Opinion Score (MOS): a native Bribri speaker rated every generated clip on a 1–5 scale (1 = bad/incomprehensible, 5 = excellent), giving 20 ratings per system plus 20 for the human reference. Because the same 20 sentences run through every system, we report both unpaired (Mann–Whitney U) and paired (Wilcoxon signed-rank) tests when comparing systems.

3 Results

Table 2 reports both metrics across conditions. On MCD, the fine-tuned VITS systems scored best, with Cabécar at 3.06 ± 1.13 dB and Spanish at 3.45 ± 1.42 dB. Spark-TTS reached 6.44 ± 1.80 dB, and the zero-shot Spanish baseline was worst at 7.18 ± 1.67 dB. All four fell under our 8 dB target threshold.

Two things stand out in the MCD scores. The first is that fine-tuning mattered far more than which base language we started from: both fine-tuned VITS models more than halved the baseline's distortion, and yet the Cabécar and Spanish results only differ by 0.39 dB, which is well within their standard deviations, so the closely related sister language

gave only a marginal edge over the distant high-resource one. The second is that Spark-TTS scored worse than either fine-tuned VITS model on MCD. As the human ratings below show, though, that does not actually mean Spark-TTS was worse in practice; the two metrics disagree, and we argue in Section 4 that the MCD gap is really an artifact of how Spark-TTS was run.

Table 2 also reports MOS, and here the human ratings reorder the systems. Spark-TTS scored highest among the trained systems at 2.75, ahead of the Cabécar VITS model at 2.55, with both Spanish conditions tied at the bottom at 2.10, while the human reference sat well above all of them at 3.85. The advantage of Spark-TTS over the Spanish fine-tune is statistically significant (Mann–Whitney U, $p = 0.006$), but its edge over the Cabécar model is not ($p = 0.18$ unpaired; $p = 0.13$ paired Wilcoxon), so we treat Spark-TTS and CJP as a tie at the top. On the human metric, however, the sister language did help within VITS, since CJP beat the Spanish fine-tune ($p = 0.03$). No system reached "good"; ratings clustered between "poor" and "acceptable," and even the human reference was rated only 3.85, so the scale ran conservative and none of these systems is yet usable in practice.

Condition	MCD ↓	MOS ↑
CJP fine-tuned	**3.06**	2.55
SPA fine-tuned	3.45	2.10
Spark-TTS	6.44	**2.75**
SPA baseline	7.18	2.10
Human reference	—	3.85

Table 2: MCD (dB, lower better) and MOS (1–5, higher better, native speaker). Bold marks the best trained system on each metric. The two metrics disagree: VITS wins on MCD, Spark-TTS on MOS.

4 Discussion

The biggest result for us is that fine-tuning mattered more than which base language we started from. Cabécar and Spanish ended up at almost the same MCD even though one is closely related to Bribri and the other really is not, and this suggests that the main bottleneck was adapting to Bribri's sounds and orthography rather than how related the base model was in the first place. That is useful to know, because if you are starting out on a new low-resource language, a high-resource model with good acoustic coverage may work about as well as a scarce sister-language checkpoint, which gives you more options for where to begin.

The VITS-versus-Spark comparison on MCD is not really apples-to-apples, and this is honestly the part of our setup we are least sure about. Our Spark-TTS inference ran with no voice prompt, so the model never actually tried to match the specific reference speaker, and because MCD counts any speaker mismatch as spectral distance, Spark-TTS was at a disadvantage on this metric no matter how good its Bribri really was. VITS, on the other hand, was trained directly toward the pooled reference voice and optimizes a spectral objective in the first place, so of course it should score well on a spectral metric. That means the 6.44 vs 3.06 dB gap is at least partly an evaluation artifact rather than a real quality difference. The MOS results bear this out: when a native speaker judged the audio directly, Spark-TTS rated as high as the best VITS system rather than worse, even though MCD had placed it far behind. So the spectral gap did not reflect a real quality gap, and we read the two metrics together rather than trusting MCD alone.

A low MCD also does not mean that the speech is actually linguistically correct, which matters a lot for a tonal language, because a sentence can be spectrally close to the reference while still getting the tone, the vowel quality, or the segments wrong. We saw this in the VITS loss-weight trade-off, where the settings that gave the lowest spectral distortion were not always the ones that sounded most intelligible, and we saw it in the metric disagreement itself, since the system with the worst MCD (Spark-TTS) was the one the native speaker rated highest. So low-resource tonal TTS should not be judged as an audio-reconstruction task alone, and a perceptual metric like MOS is necessary here rather than optional.

Ethical considerations. Bribri speech data is not a neutral dataset; it comes from a living Indigenous community and was used here under Prof. Coto-Solano's supervision, with a native speaker overseeing evaluation. A Bribri voice could support revitalization and digital accessibility, but the same capability also carries real risk, because LLM-based systems such as Spark-TTS can clone or imitate a voice fairly quickly, and that is especially sensitive for a small community whose recorded speakers are few and identifiable. Decisions about whether and how to deploy such a model should rest with Bribri speakers and educators, judged not by whether the output sounds realistic but by whether it serves community-defined goals.

5 Conclusion

We compared three transfer-learning approaches to Bribri TTS under a 25-minute data constraint: VITS fine-tuned from Cabécar, VITS fine-tuned from Spanish, and LoRA-fine-tuned Spark-TTS. By MCD the fine-tuned VITS systems scored best, but the central lesson is that fine-tuning mattered far more than the choice of base language—the close sister language offered only a marginal edge over a distant high-resource one. The two metrics also disagreed: on native-speaker MOS, Spark-TTS rated as high as the best VITS system despite its much worse MCD. We read this as confirmation that the apparent MCD advantage of VITS is an artifact of the lack of speaker conditioning in our Spark-TTS setup, so objective spectral distance alone cannot settle the architectural question here.

Future work follows directly. The clearest next step is a more controlled evaluation: speaker-conditioned Spark-TTS generation, more than one rater so MOS differences can be tested with more power, and a tone-specific intelligibility test rather than overall MOS alone. Training on a single-

speaker subset, rather than four pooled voices, would remove a known source of acoustic blur. Longer Spark-TTS fine-tuning and larger LLM-based variants are worth testing now that the pipeline runs. Most importantly, any further development should be pursued in direct collaboration with the Bribri community, so that a working voice serves revitalization, education, and accessibility on the community's terms.

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